

Brief review: vector operations in 2D

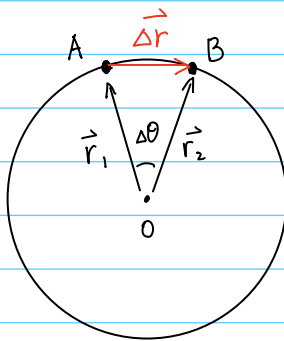
Decomposition of motion in two independent directions.

Projectile motion.

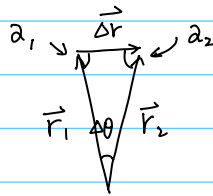
4.4 Uniform circular motion (speed is constant)

But velocity is not constant, since the direction is always changing.

velocity:



$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$$

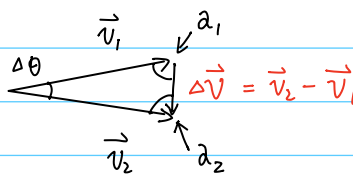
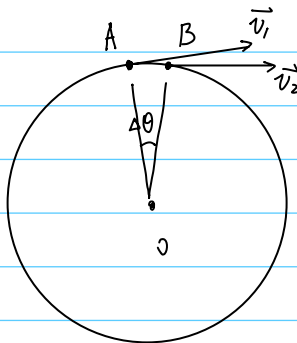


When $\Delta t \rightarrow 0$, $\Delta \theta \rightarrow 0$, $\angle \alpha_1 = \angle \alpha_2 \rightarrow 90^\circ$
that means $\Delta \vec{r}$ becomes perpendicular to $\vec{r}$

$\therefore \vec{v} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t}$ is always tangent to the circle and perpendicular to radius.

acceleration:

Since the velocity is changing, it has acceleration.



$$\Delta \vec{v} = \vec{v}_2 - \vec{v}_1$$

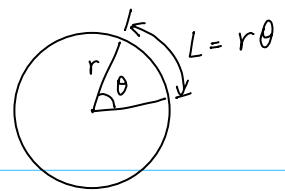
Similarly, when $\Delta t \rightarrow 0$, $\Delta \vec{v}$ becomes perpendicular to $\vec{v}$,
that is, parallel to the radius and pointing toward center

$$\vec{a} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t}$$

$\vec{a}$ is always pointing towards center.

centripetal acceleration a_c

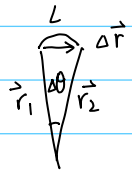
* Note that this is not a constant acceleration motion,
since the direction of $\vec{a}$ is changing.

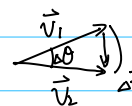


* arc length = radius $\times$ angle (in rad)

360° in rad is 2π

$$C = 2\pi r$$

In , when $\Delta t \rightarrow 0$, $\Delta r \rightarrow r\Delta\theta$ ①

In , when $\Delta t \rightarrow 0$, $\Delta v \rightarrow v\Delta\theta$ ②

$$a \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} \stackrel{②}{=} \frac{v\Delta\theta}{\Delta t} \stackrel{①}{=} \frac{v \cdot \Delta r}{r \cdot \Delta t} \stackrel{v \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta r}{\Delta t}}{=} \frac{v^2}{r}$$

centripetal acceleration magnitude $a_c = \frac{v^2}{r}$

e.g. For a fixed speed 20 m/s , driving on a curved highway ($r = 1 \text{ km}$) v.s. on a sharp turn ($r = 10 \text{ m}$).

$$a_c = \frac{20^2 \text{ m}^2/\text{s}^2}{1000 \text{ m}} = 0.4 \text{ m/s}^2 \quad a_c = \frac{20^2 \text{ m}^2/\text{s}^2}{10 \text{ m}} = 40 \text{ m/s}^2$$

period T : time duration for travelling one full circle

$$T = \frac{\text{circumference}}{\text{speed}} = \frac{2\pi r}{v} \quad (\text{SI unit: s}) \quad ①$$

frequency / rotation rate $f \equiv \frac{1}{T}$ (unit: s^{-1})

angular speed (how fast the angle rotates) since v is constant, ω is constant

$$\theta = \omega t$$

$$\omega = \frac{2\pi}{T} \quad ③$$

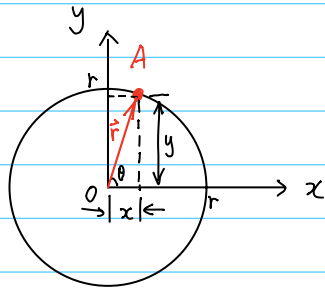
To distinguish with ω , v is called "tangential" or "linear" or "translational" speed.

From ①③: $v = r\omega$ (relation between translational speed and angular speed)

→ For a fixed angular speed, the longer the radius the higher the translational speed.

a_c can also be written as $a_c = r\omega^2$

* Describe circular motion in Cartesian coordinate.



math ref:

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

$$\theta = \omega t$$

$$x = r \cos \theta = r \cos \omega t$$

$$y = r \sin \theta = r \sin \omega t$$

$$\vec{r} = r(\cos \omega t \hat{i} + \sin \omega t \hat{j}) \quad \frac{d}{dt}$$

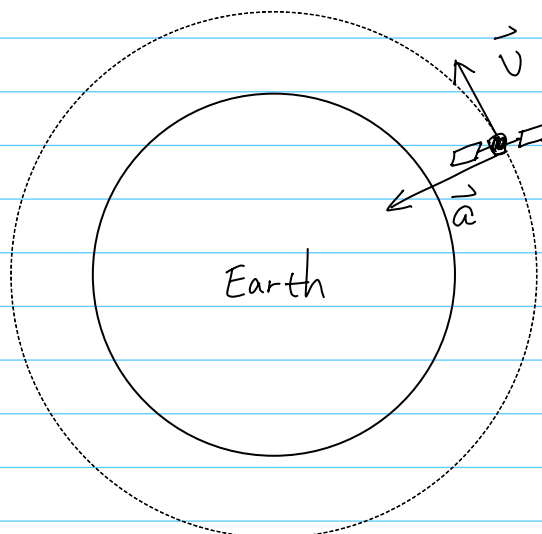
$$\vec{v} = r\omega(-\sin \omega t \hat{i} + \cos \omega t \hat{j}) \quad \frac{d}{dt}$$

$$\vec{a} = -r\omega^2(\cos \omega t \hat{i} + \sin \omega t \hat{j}) = -r\omega^2 \hat{r}$$

$\hat{r} = \frac{\vec{r}}{r}$ is unit vector

→ $\vec{a}$ is antiparallel to $\vec{r}$, pointing towards center

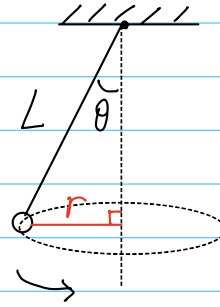
→ $|\vec{a}| = r\omega^2$ consistent with above.



For satellite:

$$|\vec{a}| \approx g$$

e.g. conical pendulum



Given L, θ , period T ,

Find $\vec{v}, \vec{a}$

Note that the circular motion radius is r , not L .

$$r = L \sin \theta$$

$$\omega = \frac{2\pi}{T}$$

$$v = \omega r = \boxed{\frac{2\pi L \sin \theta}{T}}$$

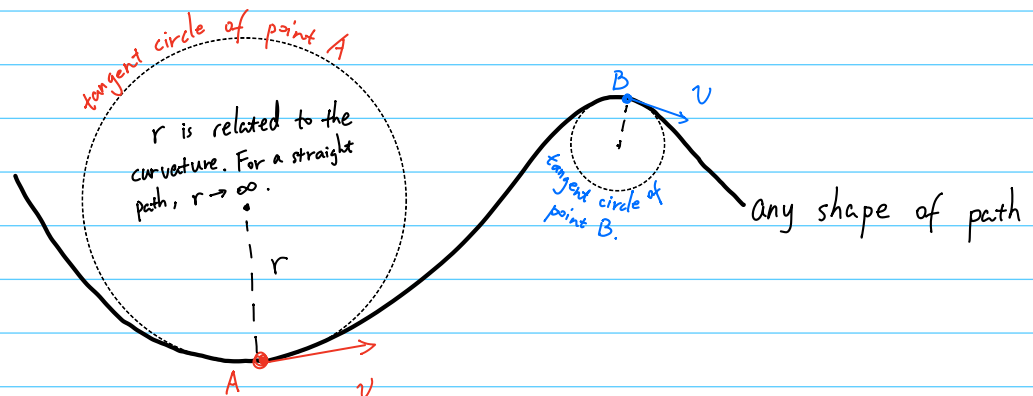
direction:
tangent to circle, pointing horizontally.

$$a = \omega^2 r = \boxed{\frac{4\pi^2}{T^2} \cdot L \sin \theta}$$

direction:
pointing horizontally towards center.
(not towards hanging point)

4.5

Tangential and radial acceleration



instantaneous acceleration can be decomposed into
a tangential component $\vec{a}_t$ and a radial component $\vec{a}_r$

$$\vec{a} = \vec{a}_t + \vec{a}_r$$

$$\vec{a}_t \perp \vec{a}_r$$

$\vec{a}_t$ = parallel or antiparallel to the instantaneous velocity $\vec{v}$
tangent to the path at that point

changes the magnitude of velocity $a_t = \frac{d|\vec{v}|}{dt}$

$\vec{a}_r$ = perpendicular to $\vec{v}$
pointing towards the center of the tangent circle

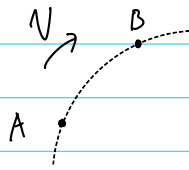
changes the direction of velocity

$$|a_r| = |a_c| = \frac{v^2}{r}$$

$\curvearrowright$ r is the radius of the tangent circle,
"radius of curvature"

* For uniform circular motion $a_t = 0$, $|a_r| = \text{constant}$
so speed constant $\swarrow$ $\searrow$ velocity changes direction.

e.g.



A car slows down steadily on a highway exit ramp.

The ramp is an arc of radius R .

The speed at point A and B is v_A and v_B , respectively.

It took t from A to B.

At the middle time between A and B find a_t , a_r , a .

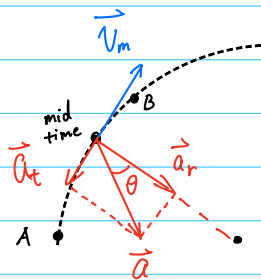
Speed reduces steadily $\Rightarrow$ reduces at a constant rate

$\Rightarrow$ instantaneous $a_t =$ average a_t from A to B

$$a_t = \left[\frac{v_B - v_A}{t} \right] < 0, \text{ pointing backwards and tangent to the circle}$$

$$\text{Speed at midtime, } v_m = \frac{v_B + v_A}{2}$$

$$a_r = \frac{v_m^2}{R} = \left[\frac{(v_B + v_A)^2}{4R} \right], \text{ pointing towards center of circle.}$$



$$a = \sqrt{a_t^2 + a_r^2}$$

$$= \sqrt{\left(\frac{v_B - v_A}{t} \right)^2 + \frac{(v_B + v_A)^4}{16 R^2}}$$

angle of a : θ backward behind the radius,

$$\text{where } \theta = \tan^{-1} \frac{a_t}{a_r} = \dots$$